



KLS
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Haliyal - 581329

PHP Programming
[BAI358D]
for
III Semester B.E.

As prescribed by
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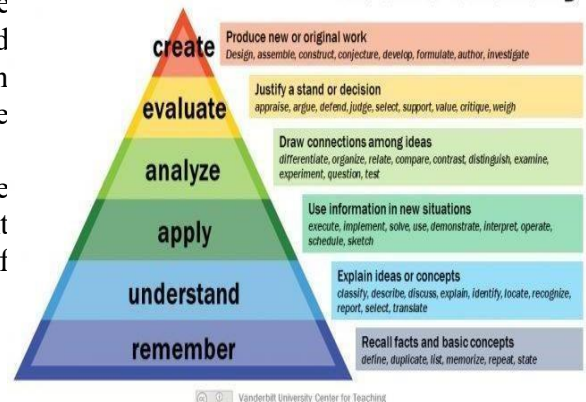
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PROGRAM OUTCOMES (POs)

Program Outcomes as defined by NBA (PO) Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Bloom's Taxonomy



Vision (College)	
To nurture talent & enrich society through excellence in technical education, research & innovation.	
Mission (College)	
To augment innovative Pedagogy & kindle quest for interdisciplinary learning & to enhance conceptual understanding.	
To build competence, professional ethics & develop entrepreneurial thinking.	
To strengthen Industry Institute Partnership & explore global collaborations.	
To inculcate culture of socially responsible citizenship.	
To focus on Holistic & Sustainable development.	
Vision (Dept)	
To lead the way in the creation and application of cutting-edge Computer science and engineering (AIML) technologies, advancing the frontiers of knowledge, and empowering future generations to drive innovation and transformation on a global scale.	
Mission (Dept)	
To train students with a strong conceptual understanding using innovative pedagogies, empowerin them to excel in the dynamic fields of Artificial Intelligence and Machine Learning.	
To imbibe professional, research, and entrepreneurial skills with a commitment to the nation's development at large.	
To strengthen the industry-institute Interaction.	
To promote life-long learning with a sense of societal & ethical responsibilities.	
Program Educational Objectives (PEO)	
PEO1	To develop an ability to identify and analyze the requirements of Computer Science and Engineering in design and providing novel engineering solutions.
PEO2	To develop abilities to work in team on multidisciplinary projects with effective communication skills, ethical qualities and leadership roles.
PEO3	To develop abilities for successful Computer Science Engineer and achieve higher career goals.
Program Specific Outcomes (PSO)	
PSO 1	To develop abilities to model real world problems using appropriate algorithms, computational theories and programming languages in the area of AI and ML..
PSO 2	To develop software applications and products in specialized areas of Artificial Intelligence and Machine Learning.

CO's And PO's Mapping Chart

Subject with code: PHP PROGRAMMING LABORATORY (BAI583D) Semester: 3rd
AY:2023-24

S.No.	Description	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
1	Apply basic concepts of PHP to develop web program	1	2			2									
2	Develop programs in PHP involving control structures	1	2	3		2									
3	Develop programs to handle structured data (object) and data items (array)	1	3			2									
4	Develop programs to access and manipulate contents of files	1	2			2									
5	Use super-global arrays and regular expressions to solve real world problems.	1	3										2		

Degree of compliance Low:1 Medium:2 High:3

Evaluation:

	Particulars	Marks	Total
CIA	Performance	20	30
	Journal	05	
	Viva-voce	05	
	Add-on Course	05	05
	Virtual Lab Experiments	05	05
	Lab IA	10	10
	Grand Total		50

Mapping of Experiments with CO, PO and PSO

S.No.	Experiment Details	CO	PO	PSO
1	a. Develop a PHP program to calculate areas of Triangle and Rectangle. b. Develop a PHP program to calculate Compound Interest.	1	1,2,5	
2	Demonstrating the various forms to concatenate multiple strings Develop program(s) to demonstrate concatenation of strings: (i) Strings represented with literals (single quote or double quote) (ii) Strings as variables (iii) Multiple strings represented with literals (single quote or double quote) and variables (iv) Strings and string variables containing single quotes as part string contents (v) Strings containing HTML segments having elements with attributes	1	1,2,5	
3	a. Develop a PHP Program(s) to check given number is: (i) Odd or even (ii) Divisible by a given number (N) (iii) Square of a another number b. Develop a PHP Program to compute the roots of a quadratic equation by accepting the coefficients. Print the appropriate messages	2	1,2,3,5	
4	a. Develop a PHP program to find the square root of a number by using the newton's algorithm. b. Develop a PHP program to generate Floyd's triangle	2	1,2,3,5	
5	a. Develop a PHP application that reads a list of numbers and calculates mean and standard deviation. b. Develop a PHP application that reads scores between 0 and 100 (possibly including both 0 and 100) and creates a histogram array whose elements contain the number of scores between 0 and 9, 10 and 19, etc. The last "box" in the histogram should include scores between 90 and 100. Use a function to generate the histogram.	3	1,2,5	
6	a. Develop PHP program to demonstrate the date() with different parameter options. b. Develop a PHP program to generate the Fibonacci series using a recursive function	3	1,2,5	
7	Develop a PHP program to accept the file and perform the following (i) Print the first N lines of a file (ii) Update/Add the content of a file	4	1,2,5	
8	Develop a PHP program to read the content of the file and print the frequency of occurrence of the word accepted by the user in the file	4	1,2,5	
9	Develop a PHP program to filter the elements of an array with key names. Sample Input Data: 1st array: ('c1' => 'Red', 'c2' => 'Green', 'c3' => 'White', 'c4' => 'Black') 2nd array: ('c2', 'c4') Output: Array ([c1] => Red	3	1,2,5	

	[c3] => White)			
10	Develop a PHP program that illustrates the concept of classes and objects by reading and printing employee data, including Emp_Name, Emp_ID, Emp_Dept, Emp_Salary, and Emp_DOJ.	3	1,2,5	
11	a. Develop a PHP program to count the occurrences of Aadhaar numbers present in a text. b. Develop a PHP program to find the occurrences of a given pattern and replace them with a text	5	1,2,12	
12	Develop a PHP program to read the contents of a HTML form and display the contents on a browser.	5	1,2,12	

INTRODUCTION TO HTML

Web site: A set of interconnected web pages, usually including a homepage, generally located on the same server, and prepared and maintained as a collection of information by a person, group, or organization.

Web Page: A web page is a document that's created in html that shows up on the internet when you type in or go to the web page's address.

Types of Web Pages:

- **Static web page:** is delivered exactly as stored, as web content in the web server's file system. Contents cannot be changed.
- **Dynamic web page:** is generated by a web application that is driven by server-side software or client-side scripting. Dynamic web pages help the browser (the client) to enhance the web page through user input to the server. Contents can be changed as evolution over time.

Client–Server Model

The client–server model is a distributed application structure in computing that partitions tasks or workloads between the providers of a resource or service, called servers, and service requesters, called clients. Often clients and servers communicate over a computer network. A server is a host that is running one or more server programs which share their resources with clients. A client requests a server's content or service function.

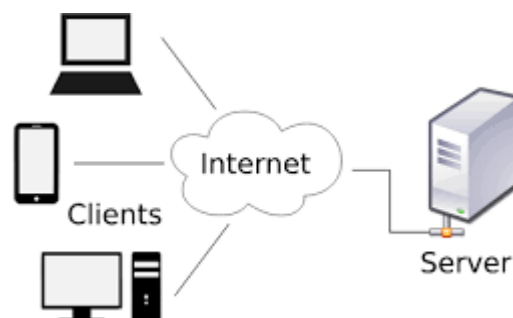


Figure 1: Client-Server Model

Web –Server

Web server refers to either the hardware (the computer) or the software (the computer application) that helps to deliver web content that can be accessed through the Internet. The most common use of web servers is to host websites, but there are other uses such as gaming, data storage or running enterprise application.

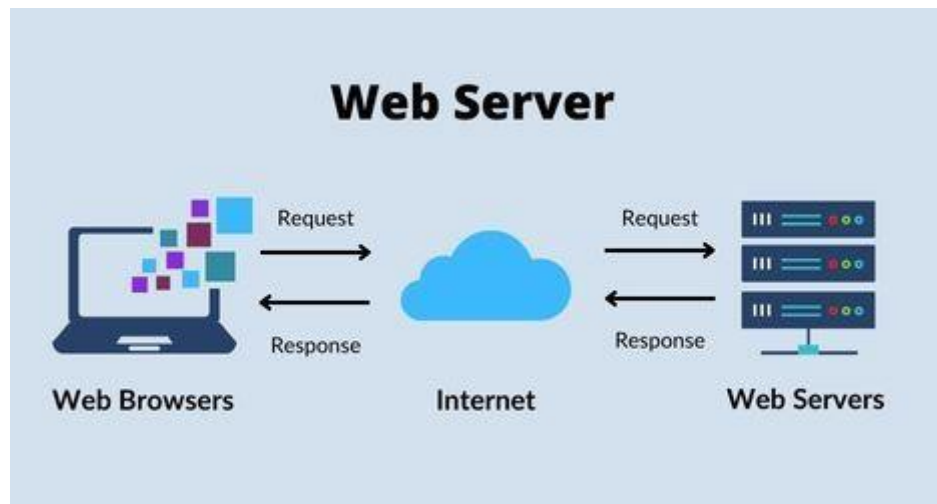


Figure 2: Web-Server

Working of different types of web pages

The different types of web pages are :

Advocacy: An advocacy web page is one sponsored by an organization to influence opinion. URL ends with .org

Business and marketing: It is one sponsored by a commercial enterprise to sell or market their services. URL ends with .com

News: It provides timely information about current events and issues.

Informational: This includes reports, research findings, schools and college information. URL ends with .edu or .gov.

Personal: It is created by an individual for his /her own personal need. URL has tittle(~).

General structure of a Web Page

A basic HTML page contains a Head section and a Body section. The contents of the head section are normally invisible in a web browser and mainly consists of some Metatags. The Body consist of those HTML elements that you want to have displayed in your browser.

```
<html>
```

```
<head>
```

```
</head>
```

```
<body>
```

```
</body>
```

```
</html>
```

Scripting language: A scripting language or script language is a programming language that supports the writing of scripts, programs written for a special runtime environment that can interpret and automate the execution of tasks which could alternatively be executed one by-one by a human operator.

URL: A uniform resource locator (URL), also known as web address, is a specific character string that constitutes a reference to a resource. In most web browsers, the URL of a web

page is displayed on top inside an address bar. An example of a typical URL would be "http://en.example.org/wiki/Main_Page".

ILLUSTRATING HTML TAGS AND THEIR ATTRIBUTES

HTML: Hyper Text Markup Language is the most widely used language to write web pages. it is a markup language.

Hypertext : Refers to the way in which web pages are linked together.

Markup Language: The user simply markups a text document with tags that tell a web browser how to structure it to display.

Creating HTML document : To begin coding HTML user needs only two things:

1. A simple text editor (notepad).
2. A web browser.

Simple steps to create a basic HTML document:

1. Open notepad or another text editor.
2. At the top of the page type <html>
3. Add the opening header tag <head>
4. On the next line type <title> give title for page </title>
5. Go to next line and type closing header tag </head>
6. Go to next line and type opening body tag <body>
7. Go to next line and type closing body tag </body>
8. Finally, go to next line and type </html>
9. In the file menu, choose save as.
10. In the save as type option box, choose all files.
11. Name the file filename.html
12. Click save.

HTML document structure: An HTML document starts and ends with <html> and </html> tags. These tags tell the browser that the entire document is composed in HTML. Inside these two tags, the document is split into 2 sections:

1. The <head>.....</head> elements contains information about the document such as title of the document etc.
2. The <body>.... </body> elements contains the real content of the document that you see on your screen.

ATTRIBUTES:

An attribute is used to define the characteristics of an element and is placed inside the element's opening tag. All attributes are made up of 2 parts: a name and a value.

-The name is the property you want to set.

-The value is what you want the value of the property to be.

Example:

INTRODUCTION TO PHP

PHP (recursive acronym for PHP: Hypertext Pre-processor) is a widely-used open source general-purpose scripting language that is especially suited for web development and can be embedded into HTML. It allows web developers to create dynamic content that interacts with databases. PHP is basically used for developing web based software applications. **PHP** is mainly focused on server-side scripting, so you can do anything any other CGI program can do, such as collect form data, generate dynamic page content, or send and receive cookies. Code is executed in servers, that is why you'll have to install a sever-like environment enabled by programs like XAMPP which is an Apache distribution.

Where is PHP used?

Three are the main areas where PHP scripts are used:

- **Server-side scripting**
This is the most used and main target for PHP. You need three things to make this work the way you need it. The PHP parser (CGI or server module), a web server and a web browser. You need to run the web server where. You can access the PHP program output with a web browser, viewing the PHP page through the server. All these can run on your home machine if you are just experimenting with PHP programming.
- **Command line scripting**
You can make a PHP script to run it without any server or browser. You only need the PHP parser to use it this way. This type of usage is ideal for scripts regularly executed using cron (on Linux) or Task Scheduler (on Windows). These scripts can also be used for simple text processing tasks.
- **Writing desktop application**
PHP may not the very best language to create a desktop application with a graphical user interface, but if you know PHP very well, and would like to use some advanced PHP features in your client-side applications you can also use PHP-GTK to write such programs. You also have the ability to write cross-platform applications this way. In this article, we'll have a detailed look at the server-side scripting using PHP.

Why PHP?

There stand convincing arguments for all those who wonder why PHP is so popular today:

- **Compatible with almost all servers used nowadays**
A web server is an information technology that processes requests via HTTP, the basic network protocol used to distribute information on the World Wide Web. There exist many types of web servers that servers use. Some of the most important and well-known are: Apache HTTP Server, IIS (Internet Information Services), lighted, Sun Java System Web Server etc. As a matter of fact, PHP is compatible with all these web servers and many more.
- **Php will run on most platforms**

Unlike some technologies that require a specific operating system or are built specifically for that, PHP is engineered to run on various platforms like Windows, Mac OSX, Linux, Unix etc)

- **PHP supports such a wide range of databases**

An important reason why PHP is so used today is also related to the various databases it supports (is compatible with). Some of these databases are: DB++, dBase, Ingres, Mongo, MaxDB, MongoDB, mSQL, Mssql, MySQL, OCI8, PostgreSQL, SQLite, SQLite3 and so on.

- **PHP is free to download and open source**

Anyone can start using PHP right now by downloading it from php.net. Millions of people are using PHP to create dynamic content and database-related applications that make for outstanding web systems. PHP is also open source, which means the original source code is made freely available and may be redistributed and modified.

- **Easy to learn & large community**

PHP is a simple language to learn step by step. This makes it easier for people to get engaged in exploring it. It also has such a huge community online that is constantly willing to help you whenever you're stuck (which actually happens quite a lot). The graphic below shows a basic workflow of dynamic content being passed to and from the client using PHP:

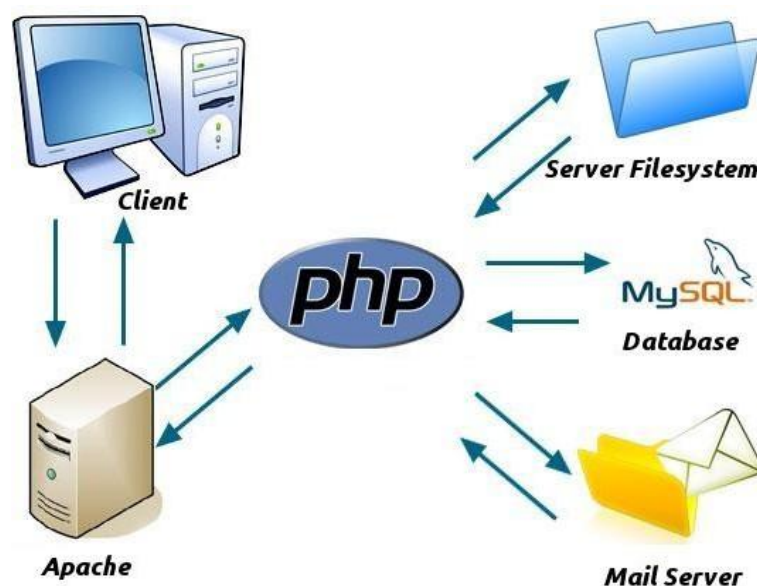


Figure 3: PHP Dynamic Content Interaction

XAMPP

Setup

XAMPP is a free and open source cross-platform web server solution developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages. In order to make your PHP code execute locally, first install XAMPP.

Download XAMPP

1. Install the program (check the technologies you want during installation)
2. Open XAMPP and click on "Start" on Apache and MySQL (when working with

databases)

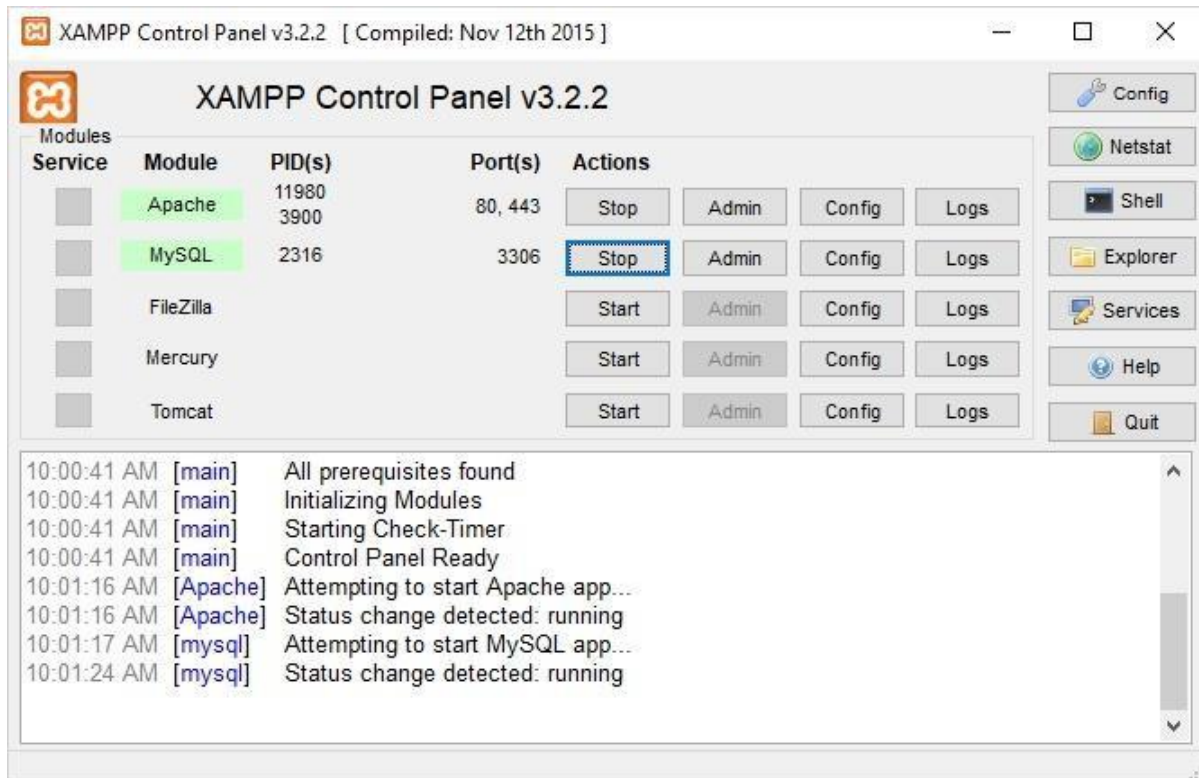


Figure 4: XAMPP window after a successful installation with Apache and MySQL enabled

Place your web project inside the htdocs directory. In the common case, if you installed XAMPP directly inside the C: drive of your PC, the path to this folder would be: C:xampphtdocs

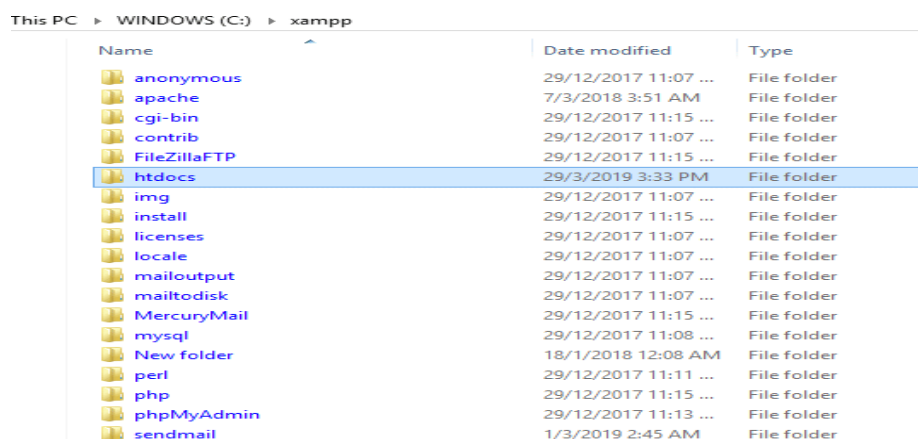


Figure 5: XAMPP Directory for Web Projects

To test the services are up and running you can just enter **localhost** in your address bar and expect the welcoming page.

PHP Language Basics

The aim of this section is to introduce the general syntax you should expect in PHP.

- Escaping to PHP

There are four ways the PHP parser engine can differentiate PHP code in a webpage:

- **Canonical PHP Tags**

This is the most popular and effective PHP tag style and looks like this:

```
<?php...?>
```

- **Short-open Tags**

These are the shortest option, but they might need a bit of configuration, and you might either choose the --enable-shorttags configuration option when building PHP, or set the short_open_tag setting in your php.ini file.

```
<?...?>
```

- **ASP-like Tags**

In order to use ASP-like tags, you'll need to set the configuration option in the php.ini file:

```
<%...%>
```

- **HTML script Tags**

You can define a new script with an attribute language like so:

```
<script language="PHP">...</script>
```

ii. Commenting PHP

Just like other languages, there are several ways to comment PHP code. Let's have a look at the most useful ones:

Use # to write single-line comments

```
<?
```

```
# this is a comment in PHP, a single line comment
```

```
?>
```

Use // to also write single-line comments

```
<?
```

```
// this is also a comment in PHP, a single line comment
```

```
?>
```

Use /* ...*/ to write multi-line comments

```
<?
```

```
/* this is a multiple line comment
```

```
Name: Web Code
```

```
Type: Website
```

```
Purpose: Web Development
```

```
*/
```

```
?>
```

PHP Data Types

Variables can store data of different types, and different data types can do different things.

PHP supports the following data types:

- String
- Integer
- Float (floating point numbers - also called double)
- Boolean
- Array

- Object
- NULL
- **PHP String**
 - A string is a sequence of characters, like "Hello world!".
 - A string can be any text inside quotes. You can use single or double quotes.
- **PHP Variable**
 - An integer data type is a non-decimal number.
 - An integer must have at least one digit.
 - An integer can be either positive or negative.
 - Integers can be specified in three formats: decimal (10-based), hexadecimal (16-based - prefixed with 0x) or octal (8-based - prefixed with 0)
- **PHP Float**
 - A float (floating point number) is a number with a decimal point or a number in exponential form.
- **PHP Boolean**
 - A Boolean represents two possible states: TRUE or FALSE.
\$x = true;
\$y = false;
Booleans are often used in conditional testing.
- **PHP Array**
 - An array stores multiple values in one single variable.
 - An array is a special variable, which can hold more than one value at a time.
 - An array can hold many values under a single name, and you can access the values by referring to an index number.
 - **Create an Array in PHP**
 - In PHP, the **array()** function is used to create an array:
array();
 - In PHP, there are **three types of arrays**:
 - **Indexed arrays** - Arrays with a numeric index
 - **Associative arrays** - Arrays with named keys
 - **Multidimensional arrays** - Arrays containing one or more arrays.
- **PHP Object**
 - An object is a data type which stores data and information on how to process that data.
 - In PHP, an object must be explicitly declared.
 - First we must declare a class of object. For this, we use the class keyword. A class is a structure that can contain properties and methods.
- **PHP NULL Value**
 - Null is a special data type which can have only one value: NULL.
 - A variable of data type NULL is a variable that has no value assigned to it.
 - If a variable is created without a value, it is automatically assigned a value of NULL.
 - Variables can also be emptied by setting the value to NULL.

PHP VARIABLE

- Variables are "containers" for storing information.
- Creating (Declaring) PHP Variables.
- In PHP, a variable starts with the \$ sign, followed by the name of the variable.
- A variable name must start with a letter or the underscore character.
- A variable name cannot start with a number.
- A variable name can only contain alpha-numeric characters and underscores (Az, 0-9, and _).
- Variable names are case-sensitive (\$age and \$AGE are two different variables).

Output Variables

- The PHP **echo** and **print** statement is often used to output data to the screen.
- **echo** has no return value while **print** has a return value of 1 so it can be used in expressions. echo can take multiple parameters (although such usage is rare) while print can take one argument. echo is marginally faster than print.

PHP Conditional Statements

In PHP we have the following conditional statements:

- **if** statement - executes some code if one condition is true.
Syntax
if (condition) {
code to be executed if condition is true;
}
- The **if... else** statement executes some code if a condition is true and another code if that condition is false.
Syntax
if (condition) {
code to be executed if condition is true;
} else {
code to be executed if condition is false;
}
- The **if....else if....else** statement executes different code for more than two conditions.
Syntax
if (condition) {
code to be executed if this condition is true;
} elseif (condition) {
code to be executed if this condition is true;
} else {
code to be executed if all conditions are false;
}
- The **switch** statement is used to perform different actions based on different conditions.
Syntax

```
switch (n) {  
    case label1:  
        code to be executed if n=label1;  
        break;  
    case label2:  
        code to be executed if n=label2;  
        break;  
    case label3:  
        code to be executed if n=label3;  
        break;  
    ...  
    default:  
        code to be executed if n is different from all labels;  
}
```

PHP Loops

- The **while** loop executes a block of code as long as the specified condition is true.

Syntax

```
while (condition is true) {  
    code to be executed;  
}
```

- The **do...while** loop will always execute the block of code once, it will then check the condition, and repeat the loop while the specified condition is true.

Syntax

```
do {  
    code to be executed; }  
while (condition is true)
```

- The **for** loops execute a block of code a specified number of times.

Syntax

```
for (init counter; test counter; increment counter) {  
    code to be executed;  
}
```

Parameters:

- **init counter:** Initialize the loop counter value
- **test counter:** Evaluated for each loop iteration. If it evaluates to TRUE, the loop continues. If it evaluates to FALSE, the loop ends.
- **increment counter:** Increases the loop counter value.
- The **foreach** loop works only on arrays, and is used to loop through each key/value pair in an array.

Syntax

```
foreach ($array as $value)  
{  
    code to be executed;
```

```
}
```

PHP Function

A user-defined function declaration starts with the word function:

Syntax

```
function functionName() {  
code to be executed;  
}
```

Function names are NOT case-sensitive.

- **Function Arguments:**
 - ✓ Information can be passed to functions through arguments. An argument is just like a variable.
 - ✓ Arguments are specified after the function name, inside the parentheses. You can add as many arguments as you want, just separate them with a comma.
- **Functions - Returning values**
 - ✓ To let a function return a value, use the return statement.

PHP File Handling

- The **fopen()** function is used to open a file. The first parameter of **fopen()** contains the name of the file to be opened and the second parameter specifies in which mode the file should be opened.
- The **fclose()** function is used to close an open file.
- The **feof()** function checks if the "end-of-file" (EOF) has been reached. The **feof()** function is useful for looping through data of unknown length.
- The **fwrite()** function is used to write to a file. The first parameter of **fwrite()** contains the name of the file to write to and the second parameter is the string to be written.

PROGRAMS

1 a. Develop a PHP program to calculate areas of Triangle and Rectangle.

```
<?php
function calculateTriangleArea($base, $height) {
    return 0.5 * $base * $height;
}
function calculateRectangleArea($length, $width) {
    return $length * $width;
}
$triangleBase = 6;
$triangleHeight = 8;
$rectangleLength = 5;
$rectangleWidth = 10;
$triangleArea = calculateTriangleArea($triangleBase, $triangleHeight);
$rectangleArea = calculateRectangleArea($rectangleLength, $rectangleWidth);
echo "Triangle Area: " . $triangleArea . " square units<br>";
echo "Rectangle Area: " . $rectangleArea . " square units<br>";
?>
```

Output

Triangle Area: 24 square units

Rectangle Area: 50 square units

1 b. Develop a PHP program to calculate Compound Interest.

```
<?php  
function calculateCompoundInterest($principal, $rate, $time, $compoundsPerYear)  
{  
    $compoundInterest = $principal * pow(1+ ($rate/ $compoundsPerYear),  
    $compoundsPerYear * $time) - $principal;  
    return $compoundInterest;  
}  
$principal = 1000;  
$rate = 0.05;  
$time = 2;  
$compoundsPerYear = 12;  
$compoundInterest=calculateCompoundInterest($principal,$rate,$time,$compoundsPerYear)  
;  
echo "Principal Amount: $" . $principal . "<br>";  
echo "Annual Interest Rate: " . ($rate * 100) . "%<br>";  
echo "Time (in years): " . $time . "<br>";  
echo "Compounds Per Year: " . $compoundsPerYear . "<br>";  
echo "Compound Interest: $" . round($compoundInterest, 2) . "<br>";  
echo "Total Amount: $" . round($principal + $compoundInterest, 2);  
?>
```

Output

Principal Amount: \$1000
Annual Interest Rate: 5%
Time (in years): 2
Compounds Per Year: 12
Compound Interest: \$104.94
Total Amount: \$1104.94

2. Develop program(s) to demonstrate concatenation of strings:**(i) Strings represented with literals (single quote or double quote)**

```
<?php
$string1 = 'Hello,';
$string2 = 'world!';
$concatenatedSingle = $string1 . $string2;
$string3 = "This is ";
$string4 = "a concatenated string.";
$concatenatedDouble = $string3 . $string4;
echo "Using Single Quotes: " . $concatenatedSingle . "\n";
echo "Using Double Quotes: " . $concatenatedDouble . "\n";
?>
```

Output

Using Single Quotes: Hello, world!

Using Double Quotes: This is a concatenated string.

(ii) Strings as variable

```
<?php
$string1 = "Hello, ";
$string2 = "world!";
$concatenatedString = $string1 . $string2;
echo $concatenatedString;
?>
```

Output

Hello, world!

(iii) Multiple strings represented with literals (single quote or double quote) and variables

```
<?php
$name = "John";
$age = 25;
$string1 = 'My name is ' . $name . ' and I am ' . $age . ' years old.';
$string2 = "My name is $name and I am $age years old.";
echo "String 1: $string1 <br>";
echo "String 2: $string2 <br>";
?>
```

Output

String 1: My name is John and I am 25 years old.
String 2: My name is John and I am 25 years old.

(iv) Strings and string variables containing single quotes as part string contents

```
<?php
$string1 = "I'm";
$string2 = "it's";
$string1 = "He said, 'Hello!'";
$string2 = 'She exclaimed, "Wow!"';
$concatenatedString1 = $string1 . ' great';
$concatenatedString2 = $string2 . ' amazing';
echo "Concatenated String 1: " . $concatenatedString1 . "<br>";
echo "Concatenated String 2: " . $concatenatedString2 . "<br>";
echo "String with single quotes 1: " . $string1 . "<br>";
echo "String with single quotes 2: " . $string2 . "<br>";
?>
```

Output

Concatenated String 1: I'm great
Concatenated String 2: it's amazing

String with single quotes 1: He said, 'Hello!'

String with single quotes 2: She exclaimed, "Wow!"

(v) Strings containing HTML segments having elements with attributes

```
<?php
$element1 = '<div class="container">';
$element2 = '<h1 id="heading">Concatenation of Strings</h1>';
$element3 = '<p style="color: blue;">Strings containing HTML segments</p>';
$element4 = '</div>';
$finalHTML = $element1 . $element2 . $element3 . $element4;
echo $finalHTML;
?>
```

Output

Concatenation of Strings

Strings containing HTML segments

3. Develop a PHP Program(s) to check given number is:**(i) Odd or even**

```
<?php
function checkOddEven($number) {
    if ($number % 2 == 0) {
        echo "{$number} is an even number.";
    } else {
        echo "{$number} is an odd number.";
    }
}
$numberToCheck = 5;
checkOddEven($numberToCheck);
?>
```

Output

5 is an odd number.

(ii) Divisible by a given number (N)

```
<?php
function isDivisible($number, $divisor) {
    if ($divisor == 0) {
        return "Cannot divide by zero!";
    }
    if ($number % $divisor == 0) {
        return "$number is divisible by $divisor";
    } else {
        return "$number is not divisible by $divisor";
    }
}
$givenNumber = 15;
$divisorNumber = 3;
$result = isDivisible($givenNumber, $divisorNumber);
```

```
echo $result;
```

```
?>
```

Output

15 is divisible by 3

(iii) Square of a another number

```
<?php
```

```
function isSquare($num) {  
    $squareRoot = sqrt($num);  
    if ($squareRoot == (int)$squareRoot) {  
        return true;  
    } else {  
        return false;  
    }  
}  
  
$numberToCheck = 16;  
if (isSquare($numberToCheck)) {  
    echo "$numberToCheck is the square of another number."  
} else {  
    echo "$numberToCheck is not the square of another number."  
}  
  
?>
```

Output

16 is the square of another number.

3 b. Develop a PHP Program to compute the roots of a quadratic equation by accepting the coefficients. Print the appropriate messages.

```
<?php  
function roots($a,$b,$c)  
{  
    $D = (($b*$b) - (4*$a*$c));  
}
```

```
if ($D >= 0){  
    $x1 = (-$b + sqrt($D))/(2*$a);  
    $x2 = (-$b - sqrt($D))/(2*$a);  
    echo "Roots are: $x1, $x2 \n";  
} else {  
    $x1 = -$b/(2*$a);  
    $x2 = sqrt(-$D)/(2*$a);  
    echo "Roots are: $x1 ± $x2 i \n";  
}  
}  
echo "Equation is x*x+5x+4=0\n";  
roots(1,5,4);  
echo "\n Equation is x*x+4x+5=0\n";  
roots(1,4,5);  
?>
```

Output

Equation is x*x+5x+4=0

Roots are: -1, -4

Equation is x*x+4x+5=0

Roots are: -2 ± 1 i

4 a. Develop a PHP program to find the square root of a number by using the newton's algorithm.

```
<?php
function squareRoot($number, $accuracy = 0.00001) {
    $x = $number;
    $epsilon = 1e-10;
    while (true) {
        $root = 0.5 * ($x + ($number / $x));
        if (abs($root - $x) < $epsilon) {
            break;
        }
        $x = $root;
    }
    return $root;
}
$number = 25;
$result = squareRoot($number);
echo "Square root of $number is: $result";
?>
```

Output

Square root of 25 is: 5

4 b. Develop a PHP program to generate Floyd's triangle.

```
<?php
function generateFloydsTriangle($rows)
{
    $number = 1;
    echo "Floyd's Triangle with $rows rows: <br>";
    for ($i = 1; $i <= $rows; $i++) {
        for ($j = 1; $j <= $i; $j++) {
```

```
        echo $number . " ";  
        $number++;  
    }  
    echo "<br>";  
}  
}  
$rows = 5;  
generateFloydsTriangle($rows);  
?>
```

Output

Floyd's Triangle with 5 rows:

```
1  
2 3  
4 5 6  
7 8 9 10  
11 12 13 14 15
```

5 a. Develop a PHP application that reads a list of numbers and calculates mean and standard deviation.

```
<?php

function findMean(&$a, $n)
{
    $sum = 0;
    for ($i = 0; $i < $n; $i++)
        $sum += $a[$i];
    return (double)$sum / (double)$n;
}

function variance( $a, $n)
{
    $sum = 0;
    for ( $i = 0; $i < $n; $i++)
        $sum += $a[$i ];
    $mean = $sum / $n;
    $sqDiff = 0;
    for ( $i = 0; $i < $n; $i++)
        $sqDiff += ($a[$i] - $mean) * ($a[$i] - $mean);
    return $sqDiff / $n;
}

function standardDeviation($a, $n)
{
    return sqrt(variance($a, $n));
}

$a = array(1, 2, 3, 4, 5);
$n = sizeof($a);
$precision = 4;
echo "Given array of elements : ";
foreach( $a as $b)
```

```

{
echo $b. " ";
}

echo " Mean = " . findMean($a, $n);
echo "Variance = ".variance($a, $n);
echo "Standard Deviation = ".number_format(standardDeviation($a, $n), $precision);
return 0;
?>

```

Output

Given array of elements : 1 2 3 4 5

Mean = 3

Variance = 2

Standard Deviation = 1.4142

5 b. Develop a PHP application that reads scores between 0 and 100 (possibly including both 0 and 100) and creates a histogram array whose elements contain the number of scores between 0 and 9, 10 and 19, etc. The last “box” in the histogram should include scores between 90 and 100. Use a function to generate the histogram.

```

<?php

function generateHistogram(array $scores) {
    $histogram = array_fill(0, 11, 0); // Initialize an array for 11 score ranges (0-9, 10-19, ..., 90-100)

    foreach ($scores as $score) {
        $index = floor($score / 10);
        $histogram[$index]++;
    }

    return $histogram;
}

$scores = [15, 25, 35, 45, 55, 65, 75, 85, 95, 100, 0, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90];
$histogram = generateHistogram($scores);
for ($i = 0; $i < count($histogram); $i++) {

```

```
$lowerBound = $i * 10;  
$supperBound = ($i == 10) ? 100 : ($i * 10 + 9);  
echo "Scores between {$lowerBound} and {$supperBound}: {$histogram[$i]}  
occurrences\n";  
}  
?>
```

Output

```
Scores between 0 and 9: 2  
occurrences Scores between 10 and 19: 2  
occurrences Scores between 20 and 29: 2  
occurrences Scores between 30 and 39: 2  
occurrences Scores between 40 and 49: 2  
occurrences Scores between 50 and 59: 2  
occurrences Scores between 60 and 69: 2  
occurrences Scores between 70 and 79: 2  
occurrences Scores between 80 and 89: 2  
occurrences Scores between 90 and 99: 2  
occurrences Scores between 100 and 100: 1
```


6 a. Develop PHP program to demonstrate the date() with different parameter options.

```
<?php

echo "<br>". "Today is".date("y/m/d"). "<br>";
echo "Today is" . date("y:m:d") . "<br>";
echo "Today is" . date("y-m-d") . "<br>";
echo "Today is" . date("l") . "<br>";
echo "Today is" . date("D") . "<br>";
echo "Today is" . date("N") . "<br>";

?>
```

Output

Today is 23/11/06
Today is 23:11:06
Today is 23-11-06
Today is Monday
Today is Mon
Today is 1

6 b. Develop a PHP program to generate the Fibonacci series using a recursive function.

```
<?php
function Fibonacci($number)
{
    if ($number == 0)
        return 0;
    else if ($number == 1)
        return 1;

    else
        return (Fibonacci($number-1) + Fibonacci($number-2));
}

$number = 10;
for ($counter = 0; $counter < $number; $counter++)
{
    echo Fibonacci($counter), ' ';
}
?>
```

Output

0 1 1 2 3 5 8 13 21 34

7. Develop a PHP program to accept the file and perform the following**(i) Print the first N lines of a file**

```
<?php

function printFirstNLines($filename, $num_lines) {

    if (!file_exists($filename) || !is_readable($filename)) {

        echo "File does not exist or is not readable.";

        return;

    }

    $file = fopen($filename, 'r');

    $count = 0;

    while (($line = fgets($file)) !== false && $count < $num_lines) {

        echo $line;

        $count++;

    }

    fclose($file);

}

$filename = "example.txt";

$num_lines = 5;

printFirstNLines($filename, $num_lines);

?>
```

Output

example.txt

Hello everyone

(ii) Update/Add the content of a file

```
<?php

function updateFileContent($fileName, $contentToAdd) {
    if (file_exists($fileName)) {
        $currentContent = file_get_contents($fileName);
        $currentContent .= $contentToAdd;
        file_put_contents($fileName, $currentContent);
        echo "Content updated/added successfully.";
    } else {
        echo "File does not exist.";
    }
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $fileToModify = $_FILES['file']['tmp_name'];
    $content = $_POST['content'];
    updateFileContent($fileToModify, $content);
}

?>

<!DOCTYPE html>

<html>

<head>

    <title>Update/Add File Content</title>

</head>

<body>

    <h1>Update/Add File Content</h1>

    <form method="post" enctype="multipart/form-data">

        <label for="file">Select File:</label>

        <input type="file" name="file" id="file"><br><br>

        <label for="content">Content to Add/Update:</label><br>

        <textarea name="content" rows="4" cols="50"></textarea><br><br>

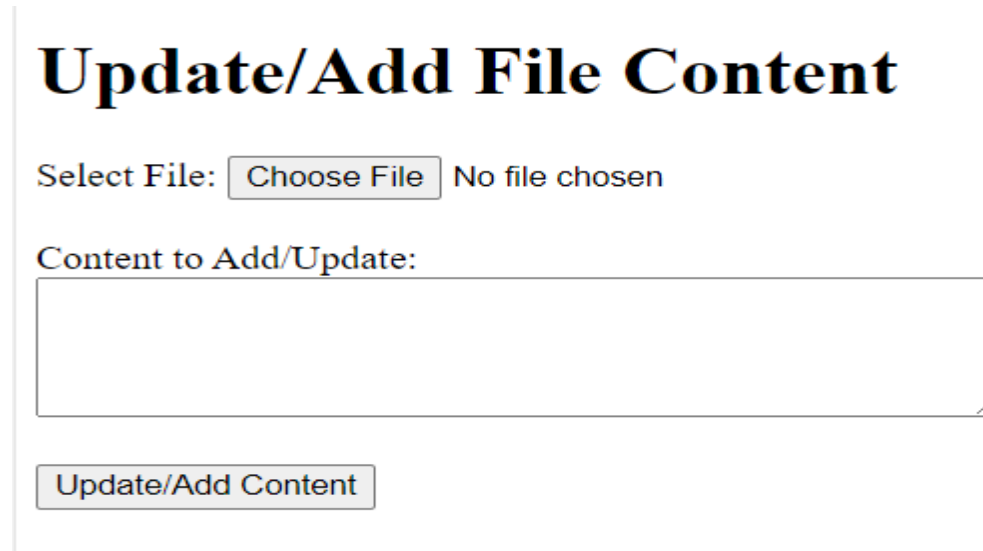
    </form>

</body>

</html>
```

```
<input type="submit" value="Update/Add Content">
</form>
</body>
</html>
```

Output



Update/Add File Content

Select File: No file chosen

Content to Add/Update:

8. Develop a PHP program to read the content of the file and print the frequency of occurrence of the word accepted by the user in the file

<?php

```
function countOccurrences($str, $word)
{
    $a = explode(" ", $str);
    $count = 0;
    for ($i = 0; $i < sizeof($a); $i++)
    {
        if ($word == $a[$i])
            $count++;
    }
    return $count;
}
$str = "GeeksforGeeks A computer science portal for geeks ";
$word = "portal";
echo (countOccurrences($str, $word));
?>
```

Output

1

9. Develop a PHP program to filter the elements of an array with key names.**Sample Input Data:****1st array: ('c1' => 'Red', 'c2' => 'Green', 'c3' => 'White', c4 => 'Black')****2nd array: ('c2', 'c4')****Output:****Array**

```
(  
[c1] => Red  
[c3] => White  
)
```

<?php

\$first_array = array('c1' => 'Red', 'c2' => 'Green', 'c3' => 'White', 'c4' => 'Black');

\$second_array = array('c1', 'c3');

print_r(array_intersect_key(\$first_array,array_flip(\$second_array)));

?>

Output**Array**

```
(  
[c1] => Red  
[c3] => White  
)
```

10. Develop a PHP program that illustrates the concept of classes and objects by reading and printing employee data, including Emp_Name, Emp_ID, Emp_Dept, Emp_Salary, and Emp_DOJ.

```
<?php

class Employee {

    public $Emp_Name;

    public $Emp_ID;

    public $Emp_Dept;

    public $Emp_Salary;

    public $Emp_DOJ;

    public function __construct($name, $id, $dept, $salary, $doj) {

        $this->Emp_Name = $name;

        $this->Emp_ID = $id;

        $this->Emp_Dept = $dept;

        $this->Emp_Salary = $salary;

        $this->Emp_DOJ = $doj;

    }

    public function displayInfo() {

        echo "Employee Information:\n";

        echo "Name: " . $this->Emp_Name . "\n";

        echo "ID: " . $this->Emp_ID . "\n";

        echo "Department: " . $this->Emp_Dept . "\n";

        echo "Salary: $" . $this->Emp_Salary . "\n";

        echo "Date of Joining: " . $this->Emp_DOJ . "\n";

    }}

$employee1 = new Employee("John Doe", "E001", "IT", 50000, "2023-01-01");

$employee2 = new Employee("Jane Smith", "E002", "HR", 45000, "2023-02-15");

$employee1->displayInfo();

echo "\n";

$employee2->displayInfo();
```

?>

Output

Employee Information:

Name: John Doe

ID: E001

Department: IT

Salary: \$50000

Date of Joining: 2023-01-01

Employee Information:

Name: Jane Smith

ID: E002

Department: HR

Salary: \$45000

Date of Joining: 2023-02-15

11 a. Develop a PHP program to count the occurrences of Aadhaar numbers present in a text.

```
<?php

function countAadhaarOccurrences($text) {

    $pattern = '/\b\d{4}\s?\d{4}\s?\d{4}\b/';

    preg_match_all($pattern, $text, $matches);

    $count = count($matches[0]);

    return $count;

}

$text = "Here are some Aadhaar numbers: 1234 5678 9012, 9876 5432 1098, 8765 4321 2109.";

$result = countAadhaarOccurrences($text);

echo "Number of Aadhaar occurrences: $result";

?>
```

Output

Number of Aadhaar occurrences: 3

11 b. Develop a PHP program to find the occurrences of a given pattern and replace them with a text.

```
<?php

function replacePattern($pattern, $replacement, $inputString) {

    $resultString = preg_replace($pattern, $replacement, $inputString);

    if ($resultString !== null)

        echo "Original String: $inputString\n";

        echo "Modified String: $resultString\n";

    } else {

        echo "Error: Unable to perform replacement.\n";

    }

}

$pattern = '/\bapple\b/i';

$replacement = 'orange';
```

```
$inputString = 'I have an Apple and a green apple in my bag.';  
replacePattern($pattern, $replacement, $inputString);  
?>
```

Output

Original String: I have an Apple and a green apple in my bag.

Modified String: I have an orange and a green orange in my bag.

12. Develop a PHP program to read the contents of a HTML form and display the contents on a browser

```
<head>

    <title>Form Reader</title>

</head>

<body>

<?php
if($_SERVER["REQUEST_METHOD"] == "POST") {

    $name = test_input($_POST["name"]);
    $email = test_input($_POST["email"]);
    $message = test_input($_POST["message"]);
    echo "<h2>Form Data:</h2>";
    echo "<p>Name: $name</p>";
    echo "<p>Email: $email</p>";
    echo "<p>Message: $message</p>";
} else {

    echo "<h2>HTML Form</h2>";

    echo '<form method="post" action="" . htmlspecialchars($_SERVER["PHP_SELF"]) . ">';
    echo '    <label for="name">Name:</label>';
    echo '    <input type="text" name="name" required><br>';
    echo '    <label for="email">Email:</label>';
    echo '    <input type="email" name="email" required><br>';
    echo '    <label for="message">Message:</label>';
    echo '    <textarea name="message" rows="4" required></textarea><br>';
    echo '    <input type="submit" value="Submit">';
    echo '</form>';
}function test_input($data) {

    $data = trim($data);

    $data = stripslashes($data);
```

```
$data = htmlspecialchars($data);  
return $data;  
}  
?>
```

</body>

</html>

Output

HTML Form

Name:

Email:

Message:

Form Data:

Name: ABC

Email: abc@klsvdit.edu.in

Message: hello everyone this is kls vdit haliyal

VIRTUAL LAB EXPERIMENT

Types of Lists in HTML

The screenshot shows the Virtual Labs website interface. The browser address bar displays the URL: `https://html-iitd.vlabs.ac.in/exp/lists-in-html/`. The page title is "Types of Lists in HTML". On the left sidebar, there is a navigation menu with links: Aim, Theory, Pretest, Procedure, Simulation, Posttest, References, and Feedback. The main content area contains the following text:

Aim

The aim of this experiment is to understand the concept of list tag and its use in HTML code.

The learner will be able to:

- Demonstrate working of lists in HTML.
- Using list in HTML pages through proper syntax.

The footer of the page includes "Community Links" (Sakshat Portal, Outreach Portal), "Contact Us" (Phone: 011-26582050, Email: support@vlabs.ac.in), and "Follow Us" (Twitter, Facebook, YouTube, LinkedIn).

Ordered list

The screenshot shows the Virtual Labs website interface for the "Types of Lists in HTML" simulation. The browser address bar displays the URL: `https://html-iitd.vlabs.ac.in/exp/lists-in-html/simulation.html`. The page title is "Types of Lists in HTML". The main content area contains a large empty box with the text "Your list will appear here". Below this box, there is a form for configuring the list:

- List: Unordered List (dropdown)
- Type: circle (dropdown)
- Item1: cse (text input)
- Item2: ece (text input)
- Item3: civil (text input)
- Item4: eee (text input)

To the right of the form, there is a code editor titled "HTML - List Tag" showing the following code:

```
<!DOCTYPE html>
<body>
  <_ type= " " >
    <li>ListItem1 </li>
    <li>ListItem2 </li>
    <li>ListItem3 </li>
    <li>ListItem4 </li>
  </_ >
</body>
</html>
```

The screenshot shows a web browser window with the URL `https://html-iitd.vlabs.ac.in/exp/lists-in-html/simulation.html`. The page title is "Types of Lists in HTML". Below the title, there is a text area labeled "Your list will appear here" which displays an ordered list:

- I. computer architechture
- II. python
- III. java
- IV. HTML

Below the list, there is a form with the following fields:

- List: --List-- (dropdown)
- Type: --Type-- (dropdown)
- Item1:
- Item2:
- Item3:
- Item4:

To the right of the form, there is a code editor titled "HTML - List Tag" showing the following code:

```
<!DOCTYPE html>
<body>
  <ul style="list-style-type: none;">
    <li>ListItem1</li>
    <li>ListItem2</li>
    <li>ListItem3</li>
    <li>ListItem4</li>
  </ul>
</body>
</html>
```

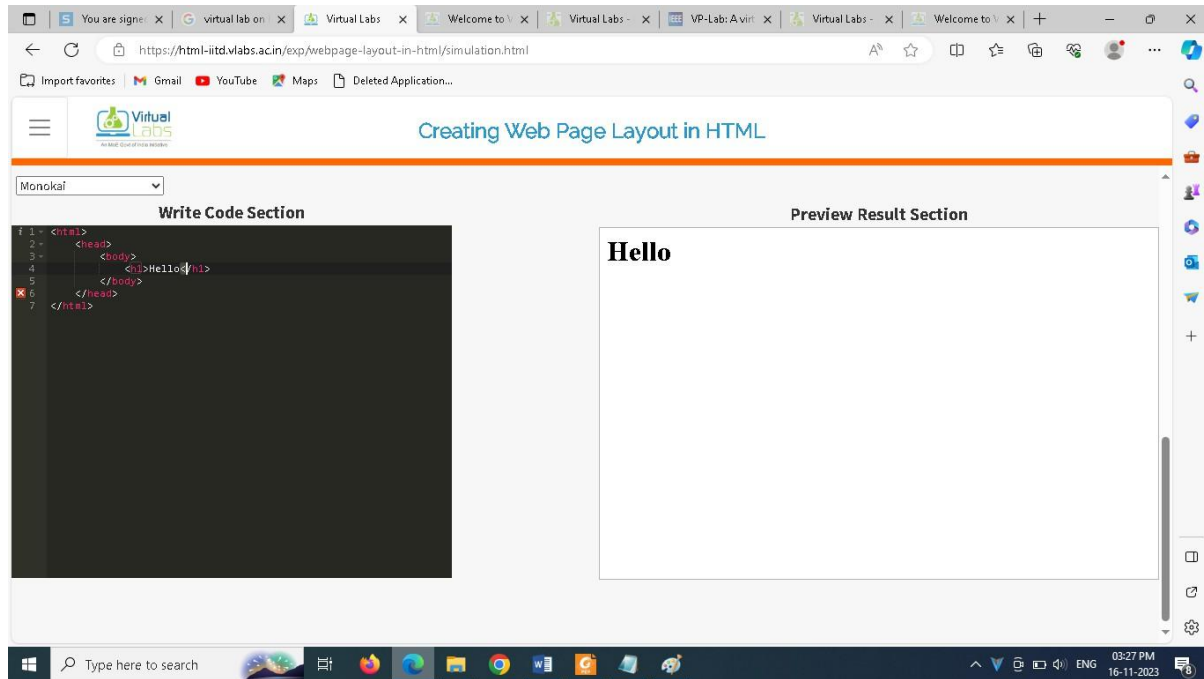
Unordered list

The screenshot shows the same web browser window as above, but the list displayed in the "Your list will appear here" area is an unordered list:

- o cse
- o ece
- o cvil
- o eee

The form fields and the code editor are the same as in the previous screenshot. The code editor shows the following code:

Creating webpage layout in HTML



Viva Question

1. What is PHP?
2. What does the initials of PHP stand for?
3. Who is the father of PHP?
4. What are the correct and the most two common way to start and finish a PHP block of code?
5. Which programming language does PHP resemble to?
6. What are the different errors in PHP?
7. What is the use of "echo" in php?
8. Differences between GET and POST methods?
9. How to declare an array in php?
10. How can PHP and HTML interact?
11. What you can do with PHP?
12. What is an efficient way to combine two variables in PHP?
13. How to create a text file in php?
14. How to find the length of a string?
15. What is the importance of "method" attribute in a html form?
16. What is the importance of "action" attribute in a html form?
17. How to find current date and time?
18. How is it possible to return a value from a function?
19. In PHP, objects are they passed by value or by reference?
20. What are the two main string operators?
21. What is the difference between for and foreach?
22. How can we change the value of a constant?
23. Is variable name case sensitive?
24. How to count the number of elements in an array?
25. Describe how PHP Works with Browser?
26. How many ways we can pass the variable through the navigation between the pages?
27. What is the difference between echo and print?
28. How can we get the browser properties using PHP?
29. How can we display the output directly to the browser?
30. What is an efficient way to combine two variables in PHP?
31. Execute a php script using command line in xampserver server in window?
32. How to find current date and time?
33. What are the method available in form submitting?
34. How to do single and multi line comment in PHP?
35. How many types of array are there in PHP?